

Experiment No. 4

OOK/ASK Communication using PlutoSDR and RTL-SDR

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Aim

To design and implement an OOK/ASK communication system using PlutoSDR as the transmitter and RTL-SDR as the receiver, and to study digital amplitude modulation and demodulation using GNU Radio.

Objective

The objective of this experiment is to study On-Off Keying (OOK) and Amplitude Shift Keying (ASK) using Software Defined Radio. The experiment demonstrates digital data transmission, carrier modulation, envelope detection, and data recovery using PlutoSDR and RTL-SDR platforms.

Theory

Amplitude Shift Keying (ASK) is a digital modulation technique in which digital information is represented by varying the amplitude of a carrier signal.

The ASK signal is represented as:

$$s(t) = A_i \cos(2\pi f_c t)$$

where A_i represents different amplitude levels corresponding to binary symbols.

On-Off Keying (OOK) is a special case of ASK where the carrier is present for binary 1 and absent for binary 0.

Binary Data	Carrier State
1	Carrier ON
0	Carrier OFF

Table 1: OOK Symbol Representation

In this experiment, random binary data modulates a carrier signal generated using GNU Radio. The modulated signal is transmitted using PlutoSDR and received using RTL-SDR. Envelope detection is then used to recover the original binary information.

Equipment Required

- ADALM-PLUTO SDR
- RTL-SDR Receiver
- GNU Radio Companion (GRC)
- Antennas
- USB Cables
- Computer/Laptop

Software Requirements

- GNU Radio
- PlutoSDR Drivers
- RTL-SDR Drivers
- libiio Support

Implemented Communication Systems

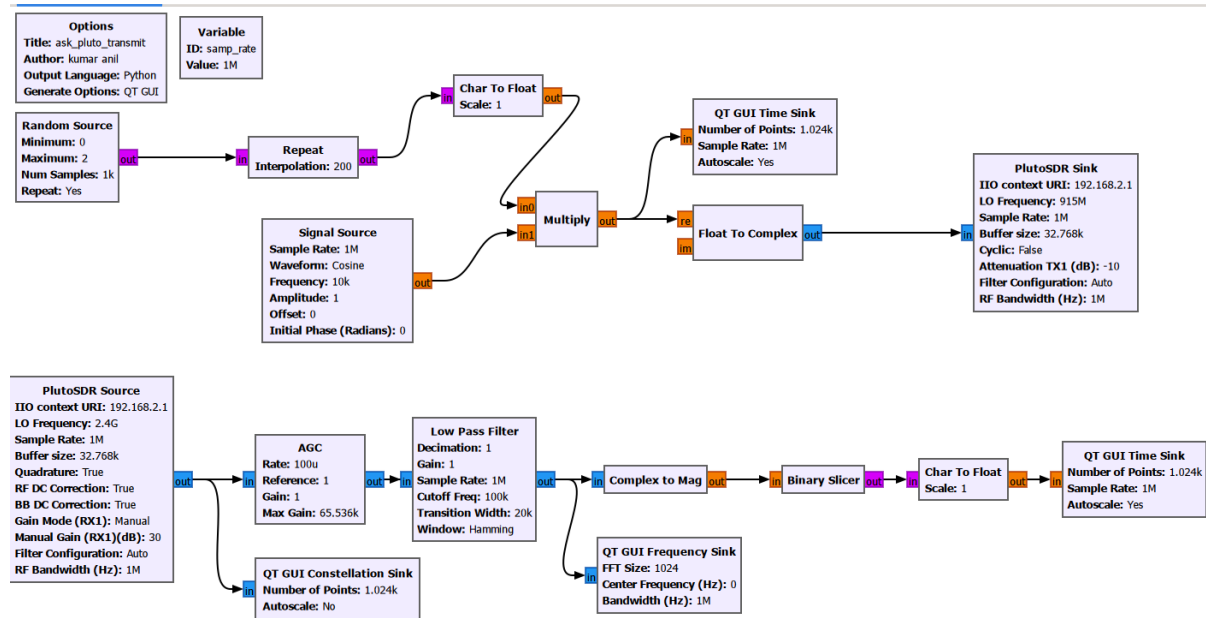


Figure 1: OOK/ASK Communication using PlutoSDR Transmitter and PlutoSDR Receiver

Description

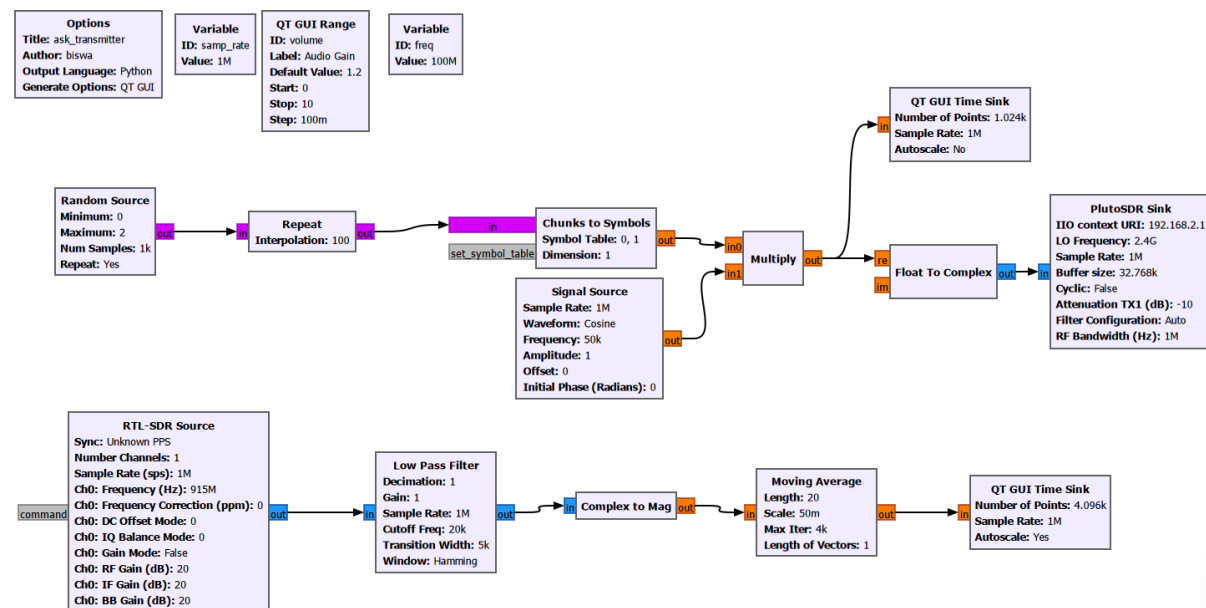


Figure 2: OOK/ASK Communication using PlutoSDR Transmitter and RTL-SDR Receiver

PlutoSDR Based OOK/ASK Communication System((Fig1

The above flowgraph represents an OOK/ASK communication system implemented entirely using PlutoSDR devices. Random binary data is generated and used to modulate a carrier signal using OOK/ASK modulation. The modulated signal is transmitted through PlutoSDR and received by another PlutoSDR receiver. The received signal is filtered and processed to recover the original binary information.

Interoperable SDR Communication System(Fig2)

The above flowgraph demonstrates interoperability between different SDR platforms. PlutoSDR is used for transmission while RTL-SDR is used for reception. The transmitted OOK/ASK signal is received by RTL-SDR, filtered, and demodulated using envelope detection techniques. This configuration verifies successful communication between heterogeneous SDR systems.

Working Principle

Random binary symbols are generated and used to control the amplitude of a carrier signal. When the transmitted bit is 1, the carrier is present. When the transmitted bit is 0, the carrier is suppressed. This creates an OOK waveform.

The RTL-SDR receiver captures the transmitted signal and performs envelope detection. The recovered amplitude corresponds directly to the transmitted binary information. Thus, the original binary sequence can be reconstructed successfully at the receiver.

Parameters

Parameter	Value
Sample Rate	1 MS/s
RF Frequency	915 MHz / 2.4 GHz
Carrier Frequency	10 kHz / 50 kHz
Modulation Type	OOK / ASK
Interpolation Factor	100 or 200
RTL-SDR Gain	20 dB
LPF Cutoff Frequency	20 kHz
LPF Transition Width	5 kHz

Table 2: System Parameters

Procedure

1. Open GNU Radio Companion.
2. Create the transmitter flowgraph using Random Source, Repeat, Chunks to Symbols, Signal Source, and Multiply blocks.
3. Configure PlutoSDR Sink for wireless transmission.
4. Create the receiver flowgraph using RTL-SDR Source.
5. Add a Low Pass Filter for noise reduction.
6. Add a Complex to Magnitude block for envelope detection.
7. Add a Moving Average block for smoothing.
8. Connect a QT GUI Time Sink to observe the recovered waveform.
9. Run both transmitter and receiver flowgraphs.
10. Observe the transmitted and recovered binary data.

Observation

Input Bit	Carrier State	Receiver Output
1	ON	High Level
0	OFF	Low Level

Table 3: Observed OOK/ASK Behaviour

The transmitted binary sequence was successfully recovered at the receiver. The presence of the carrier corresponded to logic 1, while the absence of the carrier corresponded to logic 0.

Result

An OOK/ASK communication system was successfully implemented using PlutoSDR and RTL-SDR. The transmitted binary information was recovered successfully at the receiver using envelope detection techniques.

Applications

- RFID Systems
- Remote Keyless Entry Systems
- Wireless Sensor Networks
- IoT Devices
- Low-Power Wireless Communication Systems

Conclusion

The experiment successfully demonstrated OOK/ASK communication using PlutoSDR and RTL-SDR. The transmitted binary data was recovered accurately using envelope detection, validating the operation of digital amplitude modulation techniques and interoperability between different SDR platforms.

References

1. GNU Radio Documentation
2. PlutoSDR Documentation
3. RTL-SDR Documentation
4. Digital Communication Systems by Simon Haykin